

Significant Figures and Measurement

Answer key with worked solutions

A. Understanding

1. Ruler.

The millimetre marks make 4.3 cm certain, and you can **estimate one more digit** between the marks (4.37 cm). A fourth digit (4.372 cm) would be a guess beyond what the ruler can show.

2. Accuracy and precision.

Accuracy is how close a result is to the true value; **precision** is how close repeated results are to each other. Precision describes the **spread of replicates**, so it needs at least two, and preferably three or more, measurements: one value has no spread to judge. Yes, results can be precise but not accurate: a balance that reads 0.20 g too high gives results that agree closely (small standard deviation) but are all wrong by the same amount (a systematic error).

B. Counting significant figures

3. Count.

(a) 2 (b) 4 (c) 4 (d) 5 (e) **ambiguous**: 2, 3 or 4. Write it in scientific notation to show which, e.g. 1.20×10^3 mL for 3.

C. Scientific notation and rounding

4. Scientific notation.

(a) 3.50×10^{-4} m (b) 4.56×10^4 g (c) 2.07×10^{-2} L

5. Round to 3 significant figures.

(a) **2.46 g** (b) **0.00123 m** (c) 1.88×10^4 s (18 800 s would be ambiguous) (d) **10.0 mL** (the dropped part, 51, is more than half, so 9.99 rounds up to 10.0) (e) **4.36 g** (exactly half; 6 is already even) (f) **4.38 g** (exactly half; 7 is odd, so round up to the even 8). In (c), the dropped part 50 is exactly half and 7 is odd, so 1.87 rounds to the even 1.88.

D. Calculations

6. Multiply and divide.

(a) $3.25 \text{ cm} \times 2.1 \text{ cm} = 6.825 \text{ cm}^2 \rightarrow \mathbf{6.8 \text{ cm}^2}$ (2 s.f.) (b) $100.0 \text{ g} \div 3.0 \text{ mL} = 33.33 \text{ g/mL} \rightarrow \mathbf{33 \text{ g/mL}}$ (2 s.f.) (c) $8.315 \text{ J/(mol}\cdot\text{K)} \times 298 \text{ K} = 2477.87 \text{ J/mol} \rightarrow 2.48 \times 10^3 \text{ J/mol}$ (3 s.f.; the units of K cancel)

7. Add and subtract.

(a) $12.0 \text{ g} + 0.335 \text{ g} + 4.5 \text{ g} = 16.835 \text{ g} \rightarrow \mathbf{16.8 \text{ g}}$ (1 d.p.) (b) $6.38 \text{ cm} - 2.1 \text{ cm} = 4.28 \text{ cm} \rightarrow \mathbf{4.3 \text{ cm}}$ (1 d.p.)

E. Challenge

8. Aluminium block.

$$V = 2.00 \text{ cm} \times 3.00 \text{ cm} \times 4.00 \text{ cm} = \mathbf{24.0 \text{ cm}^3}$$
 (3 s.f.)

$$\rho = \frac{m}{V} = \frac{64.8 \text{ g}}{24.0 \text{ cm}^3} = \mathbf{2.70 \text{ g/cm}^3}$$
 (3 s.f.)

9. Triplicate weighing of a 2.500 g standard.

(a) Mean: $\text{sum} = 2.52 \text{ g} + 2.48 \text{ g} + 2.51 \text{ g} = 7.51 \text{ g}$; $\bar{x} = 7.51 \text{ g} \div 3 = 2.5033 \text{ g}$ (the 3 is an exact count).

Deviations: $+0.0167 \text{ g}$, -0.0233 g , $+0.0067 \text{ g}$. Squares: $2.778 \times 10^{-4} \text{ g}^2$, $5.444 \times 10^{-4} \text{ g}^2$, $0.444 \times 10^{-4} \text{ g}^2$; $\text{sum} = 8.667 \times 10^{-4} \text{ g}^2$.

$$s = \sqrt{\frac{8.667 \times 10^{-4} \text{ g}^2}{3 - 1}} = 0.0208 \text{ g} \approx 0.021 \text{ g}$$

$$\text{RSD} = \frac{0.0208 \text{ g}}{2.5033 \text{ g}} \times 100\% = 0.83\% \text{ (the units of g cancel)}$$

$$\text{(b) Error} = 2.503 \text{ g} - 2.500 \text{ g} = \mathbf{+0.003 \text{ g}}$$
; percent error $= \frac{+0.003 \text{ g}}{2.500 \text{ g}} \times 100\% = \mathbf{+0.13\%}$

(c) The error (0.003 g) is much smaller than the standard deviation (0.021 g), so the result is **accurate**: there is no sign of a systematic error. The precision is moderate (RSD 0.8%); more replicates or a more precise balance would improve it. Result: $\mathbf{2.50 \pm 0.02 \text{ g}}$ ($\mathbf{n = 3}$).